

• Easy • #118 • Pascal's Triangle

▲ Problem Statement: Given an integer "numRows", return the first "numRows" of Pascal's Triangle.

Each number is the sum of the two numbers directly above it.

▲ Main Problem Solution: Build the triangle row by row.

- Every row starts & ends with "1"

- The middle elements are calculated using :-

$$\text{"previous}[i] + \text{previous}[i+1]\text{"}$$

- Each new row has one more element than the previous row.

▲ Brute Force Method → Calculate each element using the binomial coefficient formula.

This works, but requires extra calculations for every element & is unnecessary for this problem.

▲ Algorithm → 1. Create an empty "result".

2. Start with the first row "[1]"

3. Add the current row to "result".

4. Create the next row & add "1" at the beginning.

5. Calculate the middle elements using the previous row.

6. Add "1" in the end.

7. Repeat until "numRows" rows are created.

8. ~~Result~~ → Return "result"

▲ Important Points → First & last element of every row are always "1"

• Middle elements come from adding two adjacent elements of the previous row.

• New row size = previous row size + '1'.

• "prev = curr" makes the newly created row the previous row for the next iteration.

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▲ Complexity - Time - $O(\text{num Rows}^2)$
Space - $O(\text{num Rows}^2)$